

Car Price Prediction Using Machine Learning: A Comparative Study of Linear Regression and Random Forest

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Abstract—In this project, we tried to predict the selling price of used cars using two machine learning models: Linear Regression and Random Forest. We used a dataset of 301 used car records that included features like the car's age, present price, kilometers driven, fuel type, and transmission. Before training the models, we cleaned the data, removed unnecessary columns, and converted categorical features into numbers using one-hot encoding. We then compared both models using standard error metrics. Our results show that the Random Forest model performed much better, with an R^2 score of 0.958 compared to 0.849 for Linear Regression. We also added a simple rule to identify whether a car deal is a 'Good Deal' for the buyer. This report documents everything we did, from data preparation to final results.

Keywords—used car price prediction; machine learning; random forest; linear regression; one-hot encoding; feature engineering.

I. INTRODUCTION

Buying a used car can be tricky because it is hard to know if the price being asked is fair or not. There are many things that affect a car's price, like how old it is, how many kilometers it has been driven, what type of fuel it uses, and whether it has a manual or automatic transmission. Normally, people rely on dealers or their own guesses to figure out a fair price, which is not always accurate [1].

In this project, we tried to solve this problem using machine learning. The idea is to train a model on past car sale data so that it can learn the patterns and predict the price of any car based on its features. We used two models: Linear Regression and Random Forest. We then compared their performance to see which one works better for this kind of problem.

The rest of this report is organized as follows: Section II covers related work, Section III explains our methodology, Section IV presents our results, and Section V gives our conclusion.

II. LITERATURE REVIEW

A lot of researchers have already worked on predicting car prices using machine learning. We looked at some of their work to understand what methods work well and what challenges exist.

Wu et al. [1] used linear regression to predict car prices and found that while it gives decent results, it struggles when the relationship between features and price is not a straight line. This is a known limitation of linear models.

Pal et al. [2] tried different tree-based models including Random Forest and found that they performed much better than linear models, especially when the data has complex patterns. Random Forest does not need the data to be scaled, which makes preprocessing easier.

Gegic et al. [3] also compared multiple machine learning models for car price prediction and concluded that ensemble models like Random Forest gave the most reliable results, especially when the dataset includes both numerical and categorical features.

Richardson [4] pointed out that converting the car's manufacturing year into age (by subtracting from the

current year) is a better feature for models because age directly relates to how much a car has depreciated, unlike the raw year number.

Monburinon et al. [5] tested several regression models on a similar used car dataset and confirmed that Random Forest consistently outperformed simpler models like linear and polynomial regression in terms of accuracy.

Samruddhi and Kumar [6] specifically studied the Indian used car market and found that fuel type and transmission type are among the most important features that affect selling price, which matches our observations in this project.

Krishnaraj et al. [7] used a similar approach with gradient boosting and random forests and reported that feature importance analysis helps understand which car attributes drive the price the most, which is useful for buyers and sellers.

III. METHODOLOGY

A. Dataset

We used a dataset of 301 used car records. Each record has the following information about a car: its name, the year it was made, its current selling price (this is what we want to predict), its original showroom price, how many kilometers it has been driven, its fuel type (Petrol, Diesel, or CNG), whether it was sold by a dealer or an individual, its transmission type (Manual or Automatic), and the number of previous owners.

We first checked the dataset for any missing values. There were none, so we did not need to fill in any gaps. We also checked for duplicate rows and found none.

B. Data Preprocessing

Before we could train any model, we had to prepare the data. Here is what we did step by step:

First, we created a new feature called `Car_Age` by subtracting the car's manufacturing year from 2025. This makes more sense than using the raw year, because a car made in 2015 is 10 years old, and that is more directly related to its price than the number "2015".

$$\text{Car_Age} = 2025 - \text{Year} \quad (1)$$

Next, we removed the original Year column since Car_Age already captures the same information. We also removed the Car_Name column because there are too many unique car names, and including them would confuse the model without adding much useful information [4].

After that, we converted the categorical columns (Fuel_Type, Selling_type, and Transmission) into numbers using one-hot encoding. This creates separate binary columns for each category. We used drop_first=True to avoid a problem called the dummy variable trap, where one column can be perfectly predicted from the others, which causes issues for linear models [2].

TABLE I. DATASET FEATURES AFTER PREPROCESSING

Feature	Type	Description
Present_Price	Numeric	Showroom price
Driven_Kms	Numeric	KMs on odometer
Car_Age	Integer	2025 minus Year
Owner	Integer	No. of owners
Fuel_Type_*	Binary	One-hot encoded
Transmission_*	Binary	One-hot encoded

C. Train-Test Split

We split the data into two parts: 80% for training the model, and 20% for testing it. This is a standard practice in machine learning because we want to test our model on data it has never seen before, to get a fair idea of how well it actually works [5].

D. Models Used

We trained two models:

1) Linear Regression: This is the simplest regression model. It tries to fit a straight line through the data. We used it as a baseline to compare against.

2) Random Forest Regressor: This model builds many decision trees and averages their predictions. It is better at handling non-linear patterns in the data and is generally more accurate [2], [3].

Algorithm 1: Steps We Followed

1. Load dataset (301 records)
2. Check for nulls and duplicates
3. Create Car_Age = 2025 - Year
4. Drop Year, Car_Name columns
5. One-hot encode categorical cols
6. Split: 80% train, 20% test
7. Train Linear Regression & RF
8. Evaluate on test set
9. Derive Good Deal flag

Algorithm 1. Step-by-step process we followed in this project.

E. Evaluation Metrics

We used four standard metrics to evaluate how well our models performed:

MAE (Mean Absolute Error): The average difference between predicted and actual prices. Lower is better.

MSE (Mean Squared Error): Similar to MAE but squares the errors, so bigger mistakes are penalized more.

RMSE (Root Mean Squared Error): The square root of MSE, which brings the value back to the original unit (Lakhs).

R² Score: Tells us how much of the variation in car prices our model was able to explain. A score of 1.0 would be perfect.

IV. RESULTS & DISCUSSION

After training both models on the training data, we tested them on the 20% test set that the models had never seen. The results are shown below.

TABLE II. MODEL PERFORMANCE ON TEST SET

Model	MAE	MSE	RMSE	R ²
Linear Regression	1.216	3.481	1.866	0.849
Random Forest	0.652	0.967	0.983	0.958

A. Which Model Did Better?

Looking at the numbers in Table II, Random Forest clearly did better on every single metric. Its R^2 score of 0.958 means it was able to explain about 96% of the variation in car prices, which is pretty good. Linear Regression only got 0.849, which means it missed more patterns in the data.

The MAE of Random Forest is 0.652, which means on average its predictions were only off by about 0.65 Lakhs. Linear Regression had an MAE of 1.216, which is almost double. This happens because car prices do not always follow a straight line pattern. Things like fuel type and car age affect the price in more complicated ways that Random Forest can handle but Linear Regression cannot [3].

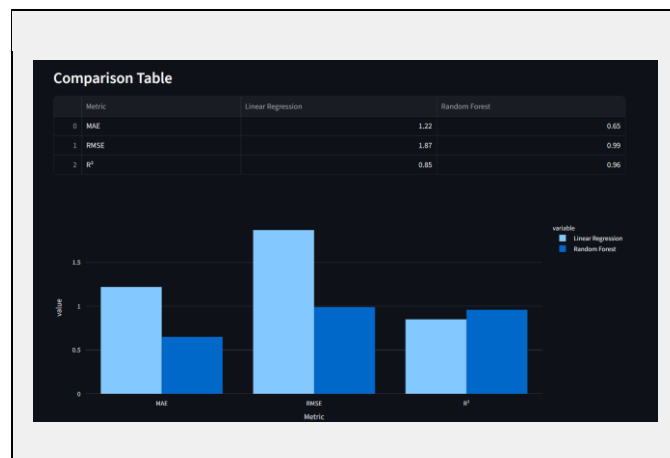


Fig. 1. Bar chart comparing MAE, RMSE, and R^2 for both models. Random Forest performs better on all three metrics.

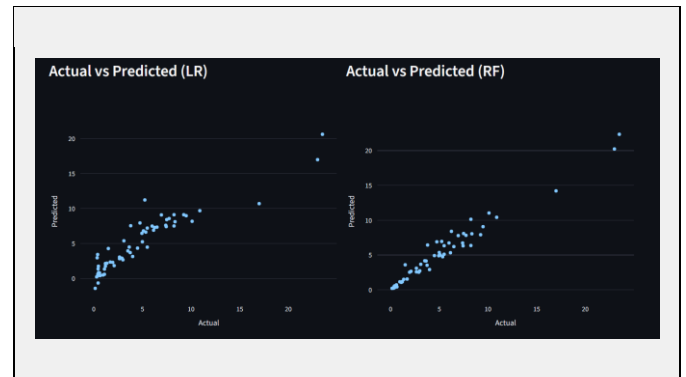


Fig. 2. Actual vs. Predicted price plots for Linear Regression (left) and Random Forest (right). Points closer to the diagonal line mean better predictions.

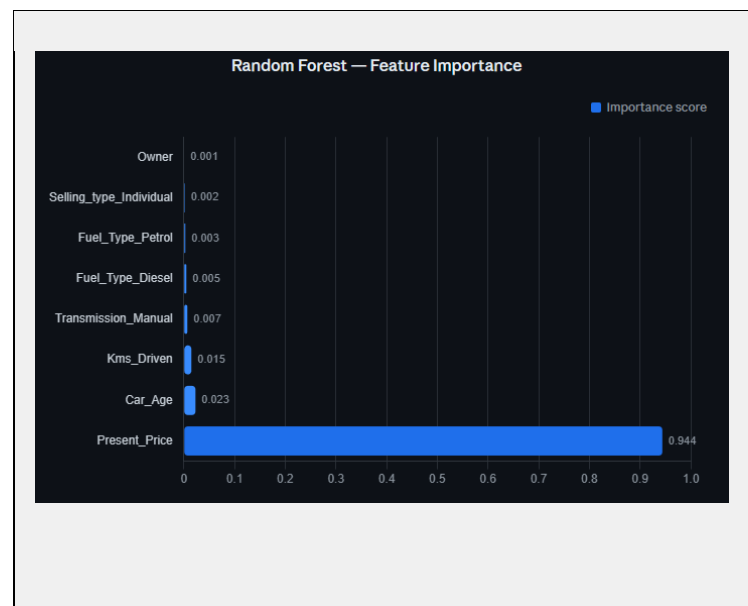


Fig. 3. Feature importance from Random Forest. Present_Price and Car_Age are the most important features for predicting price.

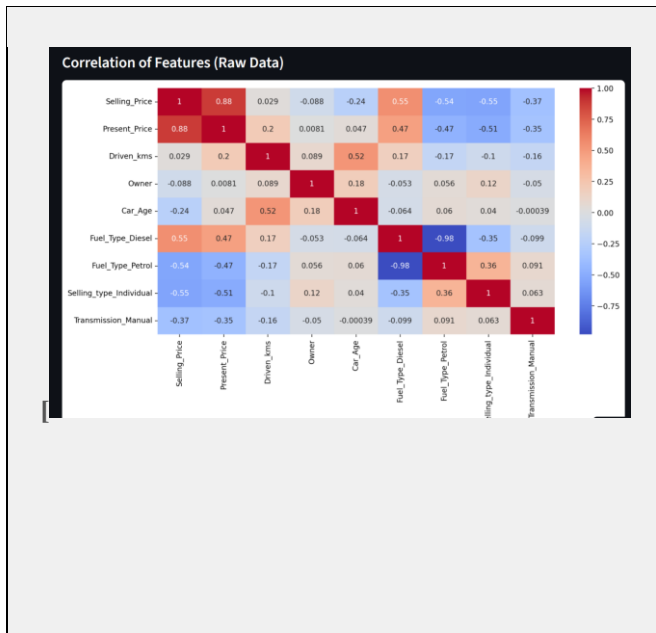


Fig. 4. Correlation heatmap showing how strongly each feature is related to Selling_Price. Present_Price has the strongest correlation (0.88).

B. Good Deal Analysis

We also added a simple feature where we check if a car is a 'Good Deal'. We defined a Good Deal as any car where the actual selling price is lower than what our model predicted. The idea is that if the model thinks the car should cost more, then buying it at the lower price is a good deal for the buyer.

Linear Regression found 175 good deals and Random Forest found 161. The reason Linear Regression finds more is because its predictions have more error and spread, so more actual prices end up below the predicted ones. This does not mean Linear Regression is better; it just means its predictions are less tight. Random Forest gives tighter predictions that are very close to the real prices, so fewer cars accidentally fall below the predicted price [6].

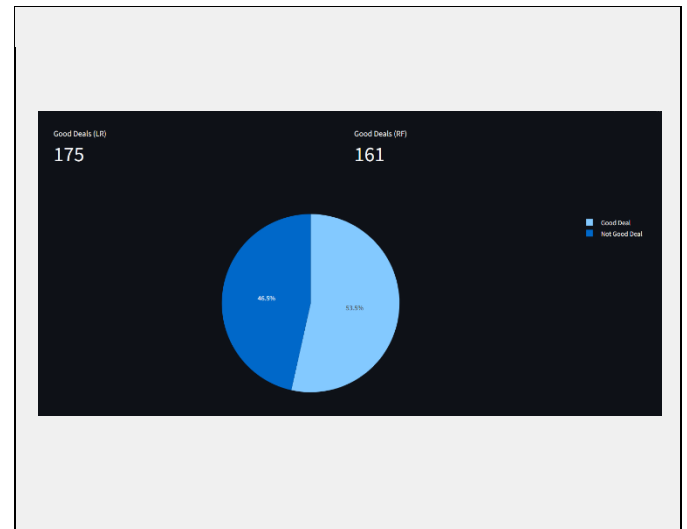


Fig. 5. Pie charts showing the percentage of Good Deal vs. Not Good Deal for both models based on our threshold rule.

V. CONCLUSION & FUTURE WORK

In this project we built a machine learning pipeline to predict used car prices and compared two models: Linear Regression and Random Forest. Based on our experiments, Random Forest was clearly the better model. It got an R^2 of 0.958 and an RMSE of 0.983, which means its predictions were quite close to the actual prices. Linear Regression was decent but not as accurate because car prices have non-linear patterns that it cannot capture.

We also learned that features like Present_Price and Car_Age are the most important for predicting car prices. Fuel type and transmission also played a role, which matches what other researchers found [6], [7].

In the future, we would like to try more advanced models like XGBoost or LightGBM to see if we can improve the accuracy even further. We could also try adding more features like the car brand or the city where it was sold. Another useful addition would be turning this into a web app where someone can enter a car's details and get an estimated price instantly.

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